DEVELOPMENT OF A SMART IOT SYSTEM FOR PRECISION AGRICULTURE

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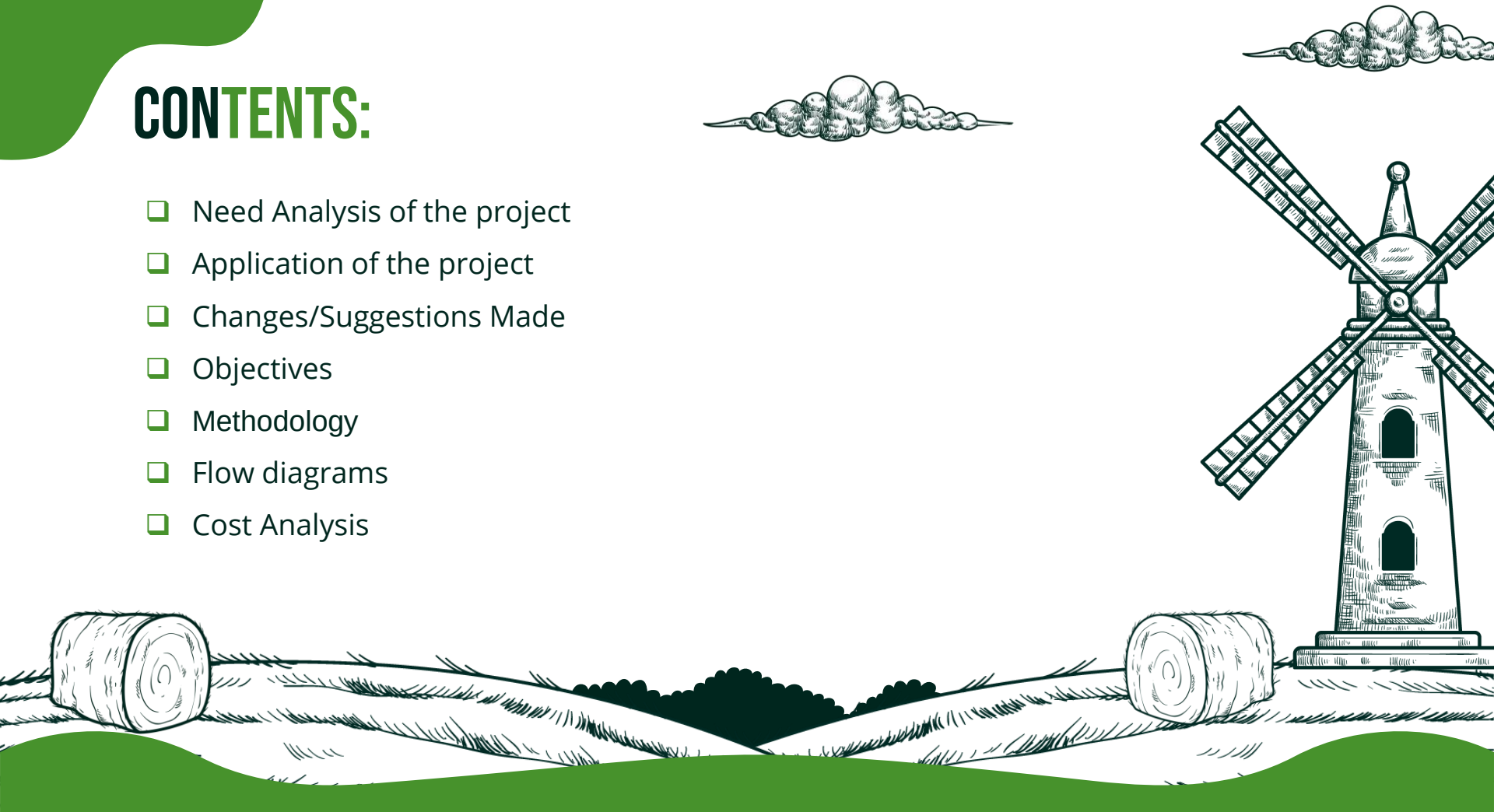
INTRODUCTION:

- Precision farming optimizes agriculture with technology, automating tasks and monitoring resources for improved efficiency.
- The capstone project pioneers agricultural innovation through a precision farming system, combining sensor tech, mesh networking, and automated controls.
- Scalability and adaptation drive the project's goal of widespread precision farming adoption, enhancing sustainability and global food security.

CONTENTS:

- Need Analysis of the project
- Application of the project
- Changes/Suggestions Made
- Objectives
- Methodology
- Flow diagrams
- Cost Analysis



NEED ANALYSIS:

- Inefficient Resource Use
- Lack of Real-time Monitoring
- Inconsistent Environmental Conditions
- Limited Control and Accessibility
- Reliance on Manual Labor
- Data Scattered Across Multiple Sources
- Risk of Crop Failure
- Environmental Impact

APPLICATION OF THE PROJECT:

- Optimizing Plant Growth
- Resource Efficiency
- Maximizing Yields
- Environmental Sustainability
- Monitoring Environmental Parameters
- Controlling Environmental Conditions
- Automation
- Data Analysis and Optimization

CHANGES/SUGGESTIONS MADE:

- Bullet points in Introduction of the Presentation
- Objectives to be of maximum 3 points
- Include Properly Labelled Block Diagram
- To control only effective components
- Design matrix for Light and Soil Moisture to have Stability and not accuracy
- Cost Analysis form properly explained

OBJECTIVES OF THE PROJECT:

- To develop sensor nodes capable of collecting data efficiently and establish an ESP mesh network to facilitate communication among sensor nodes and control nodes.
- To create a server node capable of processing sensor data and executing commands for automated control.
- To design a user-friendly dashboard for real-time monitoring and remote control of the sensor network.

METHODOLOGY

•Three types of nodes were created: sensor, server, and control nodes.

•Sensor nodes (ESP8266) collect temperature, humidity, and soil moisture data using sensors like BH1750 and DS18B20.

•Control nodes manage relays, controlling fans, humidifiers, and water pumps based on HTTP requests from the server node.

METHODOLOGY

- Server nodes (ESP32) process sensor data, send control commands, and facilitate real-time communication via Async server.

- Sensor nodes transmit data using ESP-NOW, optimized for low-latency and low-power consumption.

- The server node handles ESP-NOW requests and hosts a dashboard for real-time monitoring and device control.

FIG 1 – FLOW DIAGRAM OF PROJECT

- Sensor node collects data and sends it to the server via ESP-NOW.
- Control node adjusts relays and Neopixel based on server commands.
- Server node hosts a dashboard, updates it with sensor data, and sends control commands.
- Operates in *monitoring* (default) or *auto* mode for control adjustments.

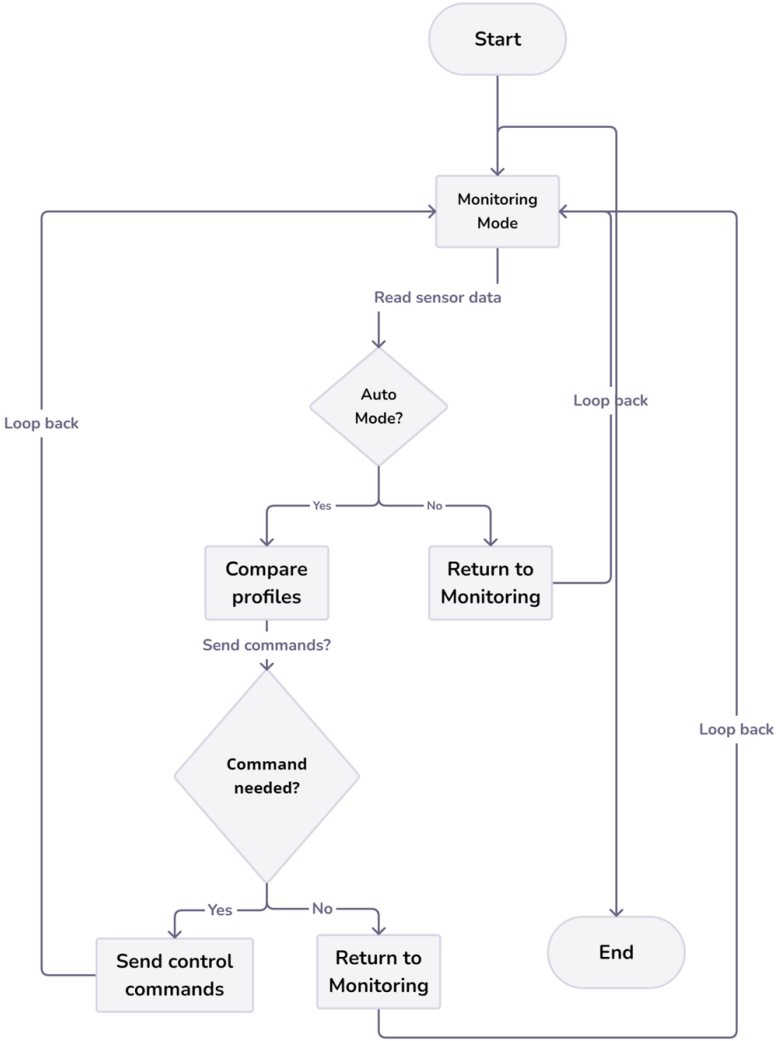
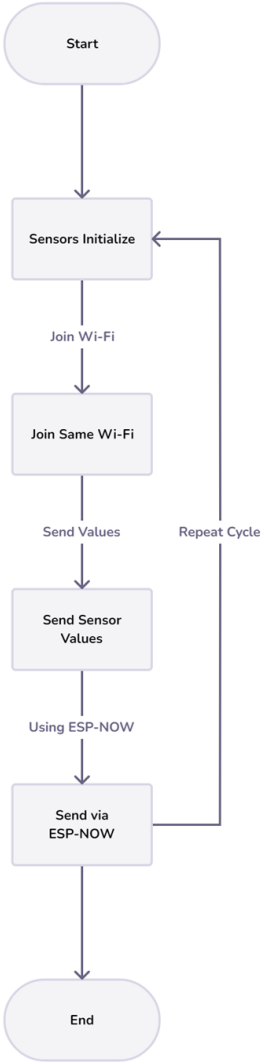
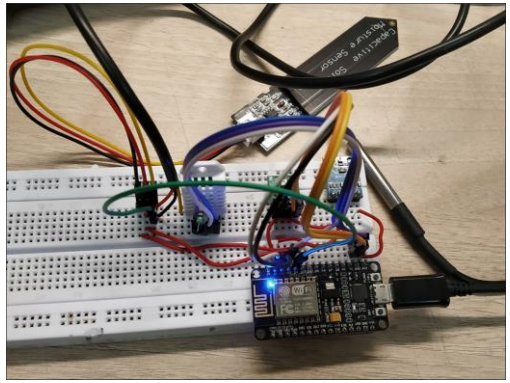


FIG 2 – FLOW DIAGRAM OF SENSOR NODE

- The sensor node initializes sensors at system start-up for data collection.
- It connects to a designated Wi-Fi channel for system communication.
- Key parameters (temperature, humidity, soil moisture) are gathered.
- Data is transmitted to the server node via ESP-NOW for real-time monitoring.

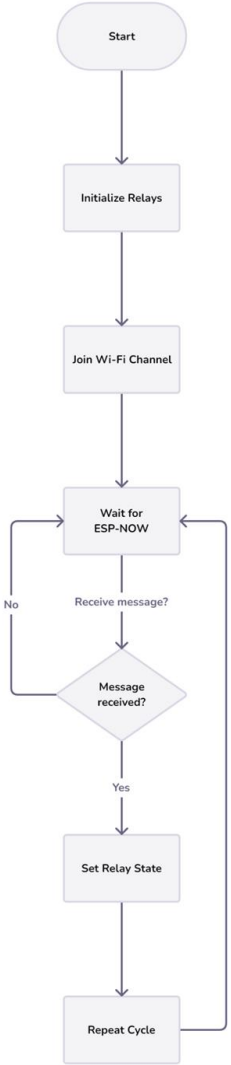


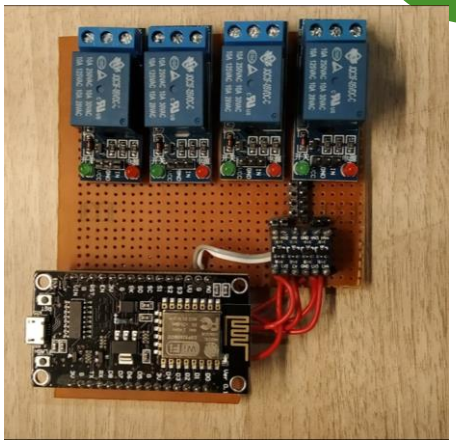


Hardware Prototype of Sensor Node

FIG 3 – FLOW DIAGRAM OF CONTROL NODE

- Control node initializes relays when the motion node is activated.
- Connects to Wi-Fi for communication with the server.
- Waits for ESP-NOW messages to adjust relays and NeoPixel as per server commands.
- Operates continuously to manage connected devices and environmental conditions.

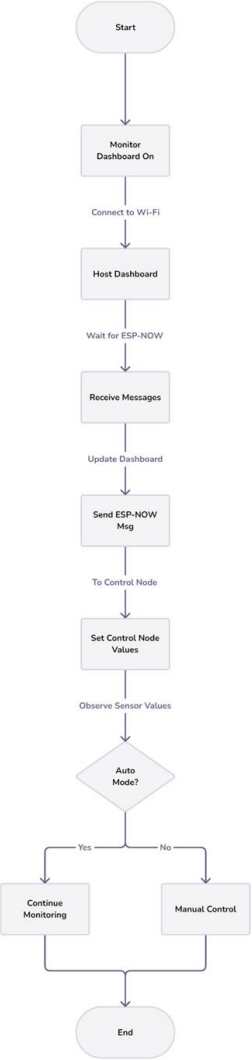


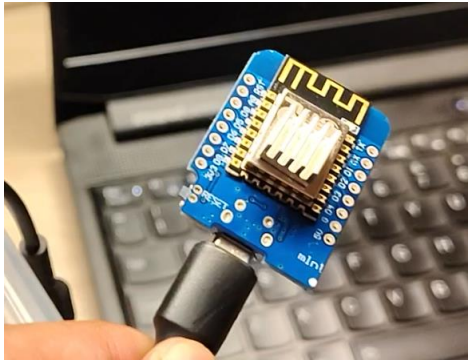


Hardware Prototype of Control Node

FIG 4 – FLOW DIAGRAM OF SERVER NODE

- Server node hosts a monitoring dashboard at server.local.
- Connects to Wi-Fi and updates the dashboard with sensor data via ESP-NOW.
- Sends control commands to the control node when needed.
- In auto mode, monitors data and adjusts controls based on thresholds.





Hardware Prototype of Server Node

COST ANALYSIS

S.No.	Name of Component	Component Description	Cost/Unit
1.	BH1750	Light sensor module	₹ 99.0/1
2.	CAPACITIVE	Capacitive soil moisture sensor	₹ 79.0/1
3.	AHT10	Temperature and humidity sensor	₹ 89.0/1
4.	DS18B20	Digital temperature sensor	₹ 67.0/1
5.	AHT21B	Temperature and humidity sensor	₹ 105.0/1
6.	DHT22	Digital temperature and humidity sensor	₹ 171.0/1
7.	WS2812B	Addressable RGB LED	₹ 35.0/1
8.	BLOWER	Air blower for ventilation	₹ 74.0/1
9.	FAN	Cooling fan	₹ 52.0/1
10.	Peltier Module	Thermoelectric cooling/heating module	₹ 550.0/1
11.	Peltier PSU	Power supply unit for Peltier module	₹ 399.0/1

COST ANALYSIS

S.No.	Name of Component	Component Description	Cost/Unit
12.	Humidifier	Humidity control device	₹ 199.0/1
13.	Water PUMP	Water pump for irrigation	₹ 39.0/1
14.	TUBE	Tubing for fluid transfer	₹ 40.0/1
15.	5V Relay	Electromechanical relay module	₹ 180.0/4
16.	Node MCU V3	Microcontroller development board	₹ 350.0/2
17.	ESP 32	Microcontroller development board	₹ 400.0/1
18.	TP4056	Lithium battery charger module	₹ 44.0/2
19.	WIRES	Electrical wires for connections	₹ 78.0/1
20.	JUMPERS	Jumper wires for prototyping	₹ 120.0/3U
21.	ADAPTER 12V	Power adapter	₹ 129.0/1
22.	HACKSAW	Cutting tool	₹ 35.0/1

COST ANALYSIS

S.No.	Name of Component	Component Description	Cost/Unit
23.	PERF BOARD	Perfboard for soldering components	₹ 39.0/1
24.	I2C EXTENSION	I2C bus extension module	₹ 79.0/1
25.	CD74HC4067	Multiplexer/demultiplexer IC	₹ 59.0/1
26.	Micro USB Cable	USB cable for data and power connection	₹ 100.0/1
27.	Level Shifter	Voltage level shifter module	₹ 199.0/1
28.	Xl6009 Step-Up	DC-DC boost converter module	₹ 65.0/1
29.	Battery 3.7V	Lithium-ion battery	₹ 380.0/2
30.	Power Cord 10A	Power cord rated for 10A current usage	₹ 40.0/1
31.	Breadboard	Reusable solderless prototyping board	₹ 56.0/1
32.	Power Supply Module	Module for converting input power	₹ 61.0/1
33.	MISC	Miscellaneous	₹ 4000.0
		Grand Total (in INR)	₹ 8373.0

THANKS!